

MGroups: A Mathematica Package for Finite Group Theory

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Summary

Group theory plays fundamental roles in both pure and applied mathematics, serving as a powerful framework for studying symmetry, structure, and transformations. Its applications span diverse areas such as algebra, number theory, geometry, physics, cryptography, and computer science among others. In recent years, computational tools have become increasingly important for exploring and experimenting with abstract algebraic structures. **MGroups** is a Mathematica package developed to support basic study of finite group structures. It offers a user-friendly interface for performing a wide range of computations involving finite groups such as cyclic groups, unit groups modulo n , dihedral groups, symmetric groups, and more. The package allows users to construct and analyse groups in terms of Cayley tables/graphs, subgroup enumeration and lattices, group homomorphisms, cosets, and related structural investigations.

Statement of need

Computational tools play a vital role in modern mathematics by automating complex and often tedious calculations, thereby freeing users from manual labour and enabling deeper exploration of abstract concepts. In educational settings, such tools are known to enhance learning by supporting experimentation, visualisation, and active engagement with abstract structures, particularly when used alongside formal theory ([Artigue, 2002](#)). Computational algebra and related mathematics analysis software have been shown to especially support pedagogy by enabling exploration, visualisation, and shifts in classroom practice beyond routine computation ([Pierce & Stacey, 2010](#)). One of the most powerful and well-established systems in discrete algebras is *GAP – Groups, Algorithms, and Programming, Version 4.15.1* ([2025](#)), which provides extensive capabilities for working with groups, rings, vector spaces, and more. GAP is widely used in research and advanced applications, offering deep algorithmic functionality and a rich library of algebraic data. However, its syntax and interface can be less accessible to beginners or those unfamiliar with command-line environments.

In contrast, MGroups is a Mathematica package developed with an emphasis on ease of use, accessibility, and pedagogical support. It is designed primarily for undergraduate students beginning with group theory, as well as for educators seeking a user-friendly tool to demonstrate group-theoretic concepts interactively. The package leverages Mathematica's intuitive syntax and symbolic computation engine to simplify many standard tasks, such as constructing Cayley tables, studying subgroups structures, exploring group morphisms, and so on. These operations can typically be performed using simple, one-line commands, greatly lowering the barrier to entry for new learners.

While tools like GAP remain the appropriate choice for large-scale or research-level computations, packages like MGroups fill a complementary niche by providing a gentle and engaging entry point to computational group theory within familiar and friendly environments.

Software design

MGroups represents finite groups as explicit data structures built from a fixed domain and a binary operation, with group properties validated at construction time. Group operations are implemented using precomputed Cayley tables and simple lookup mechanisms rather than symbolic or on-the-fly recomputation. This design ensures predictable behaviour, simplifies reasoning about group operations, and makes repeated computations efficient. The package emphasises explicitness and transparency over algorithmic optimisation, and is intentionally restricted to finite groups of small to moderate order. This choice prioritises pedagogical clarity, reproducibility, and ease of inspection within the Mathematica environment.

Research impact statement

MGroups lowers the entry barrier to computational group theory by embedding core group-theoretic computations within Mathematica's symbolic, notebook-based environment. It is well suited for institutional instruction and can be integrated into coursework to support theoretical learning through classroom demonstrations, assignments, and self-guided exploration. While not intended for large-scale or research-level computation, MGroups complements established systems by providing an accessible pedagogical tool, particularly for introductory abstract algebra courses and first-time exposure to group theory.

State of the field

A variety of computational tools for group theory are currently available, each addressing different audiences and use cases. Dedicated systems such as GAP provide extensive algorithmic depth and are widely used in research-level computations involving groups, rings, and representations. General computer algebra systems (Bosma et al., 1997; Maplesoft, 2025; Meurer et al., 2017; The Sage Developers, 2024; Wolfram Research, 2025) include group-theoretic functionality as part of a broader symbolic framework, though such functionality is often dispersed and not tailored specifically to introductory group theory. Research-oriented systems such as Magma and SageMath emphasise algorithmic depth and completeness, while general-purpose CAS such as Mathematica and Maple integrate group theory alongside many other mathematical domains. Within the Mathematica ecosystem, specialised packages such as GTPack (Geilhufe & Hergert, 2018) and GroupMath (Fonseca, 2021) extend native capabilities, with a particular emphasis on representation theory, symmetry analysis, and applications in physics. The Group Explorer (Carter & Ellis, 2019) is an excellent visualisation tool that uses pre-computed data using GAP and provides interactive representations of finite groups. Programming-language libraries, such as the group-math (Pillmore & Mullanix, 2021) package in Haskell, support abstract algebraic structures within software development workflows but are not designed as interactive educational tools. In contrast, MGroups is designed as a lightweight, pedagogy-oriented Mathematica package that focuses on core topics in finite group theory, offering a unified and accessible interface aimed at students and instructors rather than large-scale or research-oriented computation.

Installation and usage

The package is hosted at the Wolfram Paclet Repository¹ and can be installed using the single command

```
PacletInstall["Taggar/MGroups"]
```

Once installed, it can be loaded (imported) by running

¹<https://resources.wolframcloud.com/PacletRepository/resources/Taggar/MGroups/>

```
Needs["Taggar`MGroups`"]
```

Below I present some basic examples using the package

```
d = {0, 1, 2, 3, 4, 5}
f[x_, y_] := Mod[x+y, 6]
z6 = FormMGroup[d, f]
```

```
MCayleyTable[z6] (* see the Cayley table *)
Table[MCayleyGraph[z6, gen],
      {gen, {{1}, {2}, {1,2}}}] (* see Cayley graph using
                                various generating sets *)
```

```
MSubgroupLattice[MPermutationGroup[4]] (* subgroups of the
                                          group of permutations on 4 symbols *)
```

```
g = MAdditiveGroup[12]
h = {0, 6}
MFactorGroup[g, h] (* factor group g/h *)
MHomomorphism[g, g, #&] (* homomorphism from g to itself *)
```

As a special mention, the subgroup lattice and Cayley graph generated using the package are illustrated in Figure 1.

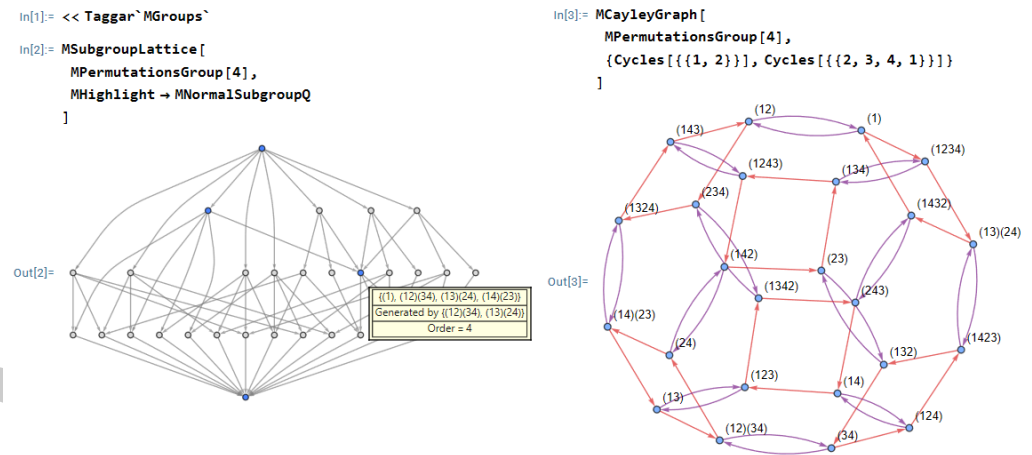


Figure 1: Subgroup lattice and Cayley graph of S_4 .

Conclusion

MGroups provides an accessible and practical entry point to computational finite group theory for Mathematica users. By prioritising clarity, ease of use, and pedagogical relevance, it complements existing research-oriented software and supports both teaching and exploratory learning in group theory.

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